

VALORANT 2026 Match Prediction Codebook

Predicting Match Outcomes in VCT 2026

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Data Source: [Valorant Champion Tour 2021–2026 Data](#) from [VLR.gg](#).

1 Overview

This codebook documents all variables in the final match-level dataset used to predict Valorant esports match outcomes. The original dataset contains 136,352 rows across 18+ source files, which were aggregated to the team-match level for modeling.

Final Dataset Dimensions: 349 match records (one row per team per match) with 19+ features per record.

2 Data Collection & Aggregation

2.1 Source Files Used

- `players_stats.csv` – Per-player performance statistics (5,231 rows)
- `overview.csv` – Team-level match overview (15,690 rows)
- `scores.csv` – Match-level results and scores (144 rows)
- `maps_scores.csv` – Team scores per map (379 rows)
- `maps_stats.csv` – Map-level statistics (412 rows)

2.2 Aggregation Method

Player-level statistics from `players_stats.csv` were aggregated to the **team-match level** by computing the mean across all 5 players on a team for each numeric performance metric. Team and match identifiers were preserved from the original data structure.

3 Variable Dictionary

3.1 Identifiers & Grouping

Variable	Type	Description	Values / Range
Tournament	String	Name of the Valorant esports tournament	e.g., “Valorant Masters Santiago 2026”
Stage	String	Tournament stage or round	e.g., “Swiss Stage”, “Group Stage”, “Playoffs”
Match Type	String	Type of match within the tournament	e.g., “Round 1”, “Round 2”, “Quarterfinals”
Teams	String	Team name(s) competing in the match	Team name (e.g., “FaZe Clan”, “Leviatán”)
Match Name	String	Human-readable match identifier	e.g., “Gentle Mates vs EDward Gaming”

3.2 Performance Metrics (Team-Level Aggregates)

All performance metrics below represent the **mean value across the 5 players on a team** for a given match.

Combat Performance

Variable	Type	Description	Values / Range	Units
Rating	Numeric	Average combat rating per player	0.0 – $\sim$ 1.5	Ratio
Average Combat Score (ACS)	Numeric	Mean combat score per player per round	0 – $\sim$ 300	Points
Kills	Numeric	Mean kills per player	0 – $\sim$ 25	Count
Deaths	Numeric	Mean deaths per player	0 – $\sim$ 25	Count
Assists	Numeric	Mean assists per player	0 – $\sim$ 15	Count
Average Damage Per Round (ADR)	Numeric	Mean damage dealt per player per round	0 – $\sim$ 200	Damage units
Headshot %	Numeric	Mean headshot percentage per player	0 – 100	Percentage
Kills Per Round (KPR)	Numeric	Mean kills per player per round	0 – $\sim$ 1.5	Kills/round
Assists Per Round (APR)	Numeric	Mean assists per player per round	0 – $\sim$ 1.0	Assists/round

First Blood Metrics

Variable	Type	Description	Values / Range	Units
First Kills (FK)	Numeric	Mean first kills per player	0 – ~10	Count
First Deaths (FD)	Numeric	Mean first deaths per player	0 – ~10	Count
Kills – Deaths (FKD)	Numeric	Mean first kill-death differential per player	–10 – ~10	Count
First Kills Per Round (FKPR)	Numeric	Mean first kills per player per round	0 – ~0.3	First kills/round
First Deaths Per Round (FDPR)	Numeric	Mean first deaths per player per round	0 – ~0.3	First deaths/round

Survival & Utility

Variable	Type	Description	Values / Range	Units
Kill, Assist, Trade, Survive %	Numeric	Percentage of rounds where player achieved a K/A/T/S	0 – 100	Percentage
Clutch Success %	Numeric	Success rate in clutch situations (1vN scenarios)	0 – 100	Percentage

Kill-Death Ratio

Variable	Type	Description	Values / Range	Units
Kills – Deaths (KD)	Numeric	Mean kill-death differential per player	–25 – ~25	Count

3.3 Match Outcome (Target Variable)

Variable	Type	Description	Values	Encoding
Match Result	Categorical	Outcome of the match for the team	“Win” or “Loss”	Binary: 1 = Win, 0 = Loss

4 Data Quality Notes

4.1 Missing Data Handling

- **Percentage columns** (Headshot %, KAST %, Clutch Success %) were cleaned by removing “%” symbols and converting to numeric format.
- Missing values in numeric columns were handled using mean imputation at the team-match aggregation level.
- Rows with excessive missing data were excluded during quality checks.

4.2 Data Validation

- Each match record represents one team’s performance in a single match.
- Matches are paired (two records per match: one for Team A, one for Team B).
- All statistical metrics are positive-valued; negative values would indicate data entry errors.
- Rating values typically range between 0.5 and 1.5 (where 1.0 is average professional level).

5 Derived Features

The following derived features may be created during model preprocessing:

Derived Feature	Calculation	Rationale
Kill Efficiency	Kills / (Kills + Deaths)	Normalized measure of elimination efficiency
First Blood Win Rate	First Kills / Total Rounds	Importance of securing first elimination
Consistency Score	ACS / Average Damage	Combat consistency relative to damage output
K/D Ratio	Kills / Deaths	Standard esports performance metric
Team Advantage (Scaled)	Team_Metric – Opponent_Metric	Feature interaction for head-to-head comparison

6 Modeling Specifications

6.1 Target Variable

- **Dependent Variable:** Match Result (Win/Loss)

- **Problem Type:** Binary Classification

6.2 Training Set

- **Sample Size:** 349 team-match records (~174–175 matches)
- **Train-Test Split:** 80/20 (standard practice)
- **Cross-Validation:** Stratified K-Fold ($k = 5$) to maintain class balance

6.3 Feature Scaling

- **Numeric Features:** StandardScaler (mean = 0, std = 1)

7 Descriptive Statistics Template

The following statistics should be computed for continuous variables during EDA:

Statistic	Interpretation
Mean	Central tendency of team performance
Median	Robust measure unaffected by outliers
Std Dev	Variability of performance metric
Min / Max	Extreme performances in dataset
Q1 / Q3	Distribution shape and quartile ranges

8 Data Collection Dates

- **Tournament Dates:** 2021–2026 Valorant Champion Tour seasons
- **Data Last Updated:** April 2026
- **Source:** VLR.gg (professional esports statistics aggregator)

9 Limitations & Assumptions

1. **Aggregation Assumption:** Individual player stats are averaged equally; star players have same weight as support players.
2. **Historical Bias:** Past tournament data may not reflect current meta, roster changes, or map updates.
3. **Sample Size:** 349 team records (~174 matches) is relatively small; models may overfit.
4. **Missing Context:** Map selection strategy, team form trends, and playoff vs. group stage dynamics not explicitly encoded.
5. **Class Imbalance:** Tournament structure (winners/losers) may create imbalanced classes.

10 File Structure

10.1 Notebook Cells

1. **Loading & Setup** – Package imports and data loading
2. **EDA & Data Cleaning** – Variable selection and exploration
3. **Aggregation** – Player-level to team-level aggregation
4. **Feature Engineering** – New feature creation
5. **Model Training** – Model selection and hyperparameter tuning
6. **Evaluation** – Performance metrics and validation

10.2 Key Output Files

- `VCTcombined_data.csv` – Raw combined dataset (136,352 rows)
- `team_match_stats.csv` – Aggregated team-match dataset (349 rows)
- `plots/` – Directory containing EDA visualizations

11 Appendix: Variable Rename Reference

Original Name	Shortened Alias	Column Type
Average Combat Score	ACS	Numeric
Kill, Assist, Trade, Survive %	KAST %	Numeric
Average Damage Per Round	ADR	Numeric
Kills Per Round	KPR	Numeric
Assists Per Round	APR	Numeric
First Kills Per Round	FKPR	Numeric
First Deaths Per Round	FDPR	Numeric
Kills – Deaths (KD)	K-D	Numeric
Kills – Deaths (FKD)	FK-D	Numeric